26900227

Discipline Specific Core Course (DSC)-MG5DSCFIM300 – Gym Instructor Essentials

MGU-UGP (Honours) | BPES | Fifth Semester | October 2026

ANSWER KEY

Part A – Questions and Answers

Answer all questions. Each question carries 1 mark.

1. **Fitness programmes for older adults should primarily emphasize what?**

Functional capacity.

2. **Heart rate during gym-based exercise can be monitored using what?**

Heart-rate monitor.

3. **A programme includes 3 sets of squats, 3 sets of lunges, and 2 sets of leg curls. What is the total number of sets?**

8 sets.

4. **Regular physical activity contributes most directly to what?**

Improved health.

5. **Internal load is primarily reflected by the body's response to what?**

Training stimulus.

6. **A trainer increases training frequency from 3 to 4 sessions per week. Which training-load factor has changed?**

Frequency.

7. **What is an important emergency safety measure in a gym?**

Knowing the location of first-aid equipment and emergency exits.

8. **What is the primary purpose of CPR?**

To maintain circulation and oxygenation.

9. **Which document is useful for organizing participants and recording test results?** Attendance/testing schedule sheet.

10. **Which instrument is used to measure blood pressure?**

Sphygmomanometer.

Part B – Short Answer Questions

Answer any TEN questions. Each question carries 2 marks.

11. Why is hydration important throughout life?

- Maintains body temperature and fluid balance.
- Supports circulation, digestion, and normal organ function.

12. Explain the importance of coordination in skilled movements.

- Enables smooth, accurate, and efficient movements.
- Improves timing, balance, and movement control.

13. What are micronutrients? Give examples.

- Micronutrients are vitamins and minerals required in small quantities for normal body functions.
- Examples include vitamins A, C, and D, and minerals such as iron, calcium, and iodine.

14. How does physical fitness influence the performance of daily activities?

- Improves strength, endurance, flexibility, and balance.
- Reduces fatigue and makes daily activities easier.

15. How can a fitness trainer improve client motivation?

- Set realistic goals and monitor progress.
- Provide encouragement, constructive feedback, and individualized programmes.

16. How can a gym instructor apply overload when resistance cannot be increased safely?

- Gradually increases repetitions or sets while maintaining proper technique.
- Modify exercise tempo, range of motion, or exercise variation appropriately.

17. What is hydration? Why is it important during exercise?

- Hydration is the maintenance of adequate fluid levels in the body.
- It supports temperature regulation, circulation, and muscle function during exercise.

18. What is progressive loading?

- Progressive loading is the gradual increase in training demand as the body adapts.
- It improves strength and fitness while reducing the risk of excessive strain.

19. What is a macrocycle?

- A macrocycle is a long-term training plan covering several months or an entire training year.
- It organizes training phases to achieve specific fitness and performance goals.

20. A conscious casualty is frightened and confused after an injury. How should a first aider respond?

- Remain calm, reassure the casualty, and explain what is happening.
- Assess the condition, seek medical assistance when necessary, and monitor the casualty.

21. What is total training load?

- Total training load is the overall demand placed on the body through training.
- It includes frequency, intensity, duration, volume, and the body's response to exercise.

22. Define flexibility as a health-related fitness component.

- Flexibility is the ability of a joint to move through its available range of motion.
- It supports mobility and efficient performance of functional movements.

Part C – Short Essay Questions

Answer any FOUR questions. Each question carries 5 marks.

23. Explain the importance of recovery capacity when applying progressive overload.

- Recovery capacity is the body's ability to repair itself and adapt after exercise.
- Adequate recovery supports muscle repair and replenishment of energy stores.

- Training load should be increased according to individual fitness and recovery ability.
- Insufficient recovery may cause excessive fatigue, reduced performance, and injury.
- Adequate sleep, nutrition, hydration, and rest days support training adaptations.

24. A trainee sustains an acute injury during exercise. Explain the response required to ensure the safety of the casualty and others.

- Stop the exercise immediately and ensure the surrounding area is safe.
- Assess the casualty's responsiveness, breathing, and severity of injury.
- Call emergency services for serious injuries and provide first aid within the limits of one's training.
- Avoid unnecessary movement, especially if a spinal injury is suspected.
- Reassure and monitor the casualty while keeping other participants away from the injured area.

25. Explain the functions and sources of iodine and the disorders caused by its deficiency.

- Iodine is an essential mineral required for the production of thyroid hormones.
- Thyroid hormones regulate metabolism, growth, and brain development.
- Sources include iodized salt, seafood, fish, dairy products, and eggs.
- Iodine deficiency may cause goitre and hypothyroidism.
- Severe deficiency during pregnancy and early childhood may impair brain development and physical growth.

26. Why can the same total number of repetitions be distributed across sets in different ways, and how does this affect training demand?

- The same total repetitions can be divided differently, such as 30 repetitions performed as 3 × 10 or 5 × 6.
- Fewer repetitions per set may help maintain proper technique and control fatigue within each set.
- More repetitions per set may increase local muscular fatigue.

- Shorter rest intervals may increase fatigue, while longer rest intervals may improve performance in subsequent sets.
- Training demand depends on resistance, number of sets and repetitions, rest intervals, and individual goals.

27. What is progression? Explain how it is applied when planning an exercise programme.

- Progression is the gradual increase in training demand as the body adapts to exercise.
- Assess the individual's health, fitness level, experience, and exercise technique.
- Gradually increase resistance, repetitions, sets, duration, frequency, or exercise difficulty.
- Individualize the programme according to the person's goals, ability, and response to training.
- Monitor progress and provide adequate recovery to improve fitness and minimize injury risk.

28. Explain the functions of fats and differentiate between saturated and unsaturated fats.

- Fats provide approximately 9 kcal of energy per gram and serve as an energy reserve.
- They insulate the body, protect internal organs, and support the absorption of vitamins A, D, E, and K.
- Saturated fats: Commonly found in butter, fatty meat, and coconut oil.
- Unsaturated fats: Found in nuts, seeds, vegetable oils, and fatty fish.
- Replacing saturated fats with unsaturated fats can help support cardiovascular health.